

MyFind Protokoll

Programm Execution

`./myfind -Ri <searchDirectory> <filename1> <filename2> ...`

searchDirectory: Path in which the search will be done

filename: Multiple files can be named to be searched (atleast 1)

-R: Recursive Search

-i: Case Insensitive

Parallelization Concepts

- for every filename in the program call, a child process will be created using forks.
- Then there are multiple If Checks
 - `pid < 0` = Error, the fork failed, stop program
 - `pid == 0` = Child Process, start Search
 - else, so `pid > 0` → this is the parent process, pid will be stored in vector
- After the for loop, so when all child processes have been started, the parent process will wait for every child pid that has been stored in the vector, to prevent zombie processes
- To ensure a correct output of the search results and prevent race conditions the output strings will first be assembled and then be written into console with one atomic write execution.
- The program also outputs the pid of the children when the search has been successful to demonstrate the parallelization

Search Function

- The search uses the directory iterator (and recursive directory iterator) to iterate through the filesystem (recursively or not, depending on how the program was executed)
- For every file the namesMatch function will be called. This function will compare the names of the searched and found file (for every file)
- If the option caseInsensitive has been selected, the two name strings will first be converted to all lowercase, to ensure case insensitivity
- If the names match, true will be returned, and the found path will be output in the console.

AI was used for the following points:

- Directory Iterator for sequencing through the file system.